# How Should We Read ViT Attention Maps? A Fast-Weight Perspective

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## Abstract

What can a bright background patch in a vision transformer's attention map tell us? Holding the model and image fixed, we changed only how we drew the map. In both DeiT checkpoints, weighting received attention by each patch's own projected value length improved alignment with foreground masks and human annotations; deletion gains stayed below our practical threshold. On pet images, assigning the same lengths to other patches removed the enrichment gain in DeiT-S. Moving scores to decoded positions lowered all three metrics in all three models. Restricting the readout to the final classification query better ranked classifier interventions in both DeiTs, but foreground AUROC fell in DeiT-S and rose in DeiT-B.

## 1 Introduction

Deep-layer attention can highlight background patches. Adding register tokens can absorb some unusually high-norm background activations [11]. What changes if we keep the image and model fixed and change only how we read the map?

For a fixed image, each attention head acts as a two-layer fast-weight network with one hidden unit per source token [31]. For one head, the attention coefficient tells us how strongly a query reads a source. Multiplying that coefficient by the length of the source's projected value gives the length of the vector it returns to that query (Fig. 1). Norm-based analyses already distinguish these quantities [19, 36]. Whether that vector length predicts foreground, human annotations or classifier sensitivity is an empirical question.

Drawing on hidden-unit analysis [4], we ask three questions about a bright patch: Is it distinctive to the image? Does its position reflect where the token's content came from? How large a vector does it return? Matched controls test the readout each question suggests. We then ask whether the gain from value weighting survives moving the lengths to other patches, and whether a map for the classifier should pool the same queries as a foreground map.

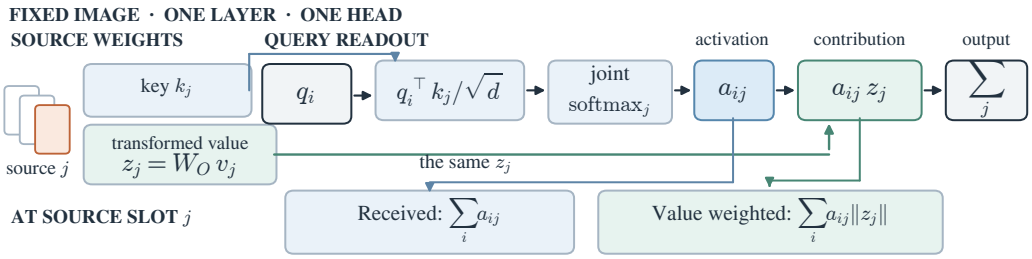



**Figure 1:** One head for a fixed image. Activation $a_{ij}$ scales source $j$'s output weight $z_j$. The two readouts sum either activation or the magnitude of the resulting vector (Eq. 1).

Preprint.

## 2 Related work

**Activation and the returned vector.** In a ViT [13], attention sets the coefficient of a source's value vector. Fast-weight memory [27, 28], Hopfield retrieval [26] and MiTA's two-layer representation [31] describe input-dependent weights and their readout. Norm-based analyses already distinguish attention coefficients from output magnitudes in language models [19] and visual attribution [36]. We ask what happens to foreground rankings when the same lengths are assigned to different patches.

**Position and information origin.** Token identity [7] and grid information [25] can survive contextual mixing. Rollout and relevance propagation trace paths across layers [1, 9, 10]; GlobEnc, ALTI and TokenTM account for additional transformations [15, 22, 34]. We test a simpler placement rule: assign a score to decoded grid positions. Registers and no-op heads help explain some high-norm outliers [6, 11, 18], whereas spatial priors can produce recurring saliency [20]. These alternatives motivate different controls.

**What should a map explain?** Attention can agree with annotations without explaining a prediction [17, 32]. Filtering aims to improve map agreement [3]; CAM targets a class score [35], and Value Zeroing measures contextual mixing [23]. Parameter dependence, supervision and occlusion choices also affect evaluation [2, 5, 30]. We evaluate the same rankings against masks and an intervention on the classifier path.

## 3 Methods for reading a frozen attention head

For query $i$, source $j$ and head $h$, let $z_j^h = W_O^h v_j^h$ be the transformed value. With key dimension $d$, the attention output before residual addition is

$$o_i = b_O + \sum_{h,j} a_{ij}^h z_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j(q_i^{h\top} k_j^h / \sqrt{d}). \tag{1}$$

For a fixed image, $k_j^h/\sqrt{d}$ and $z_j^h$ act as the input and output weights of a two-layer network in $q_i^h$, with the softmax coupling its activations across all sources. Received attention assigns $s_j = \sum_{h,i} a_{ij}^h$ to patch $j$. Summing over all queries, including CLS, gives one score per patch on the 196-patch grid.

**Specificity.** A bright patch may reflect a recurring spatial pattern. We estimate a mean map on separate calibration images, measure the spatial variance it explains on test images, and subtract it from each test map. Subtracting a centred Gaussian scaled to the mean map's spatial standard deviation controls for a radial prior.

**Location.** After several layers, a token contains information from other patches. We train a linear probe to decode its grid index and redistribute $s_j$ over the probe's predicted positions. A blur with the same mean squared displacement tests whether this outperforms smoothing alone. The probe decodes a grid index. Its accuracy tests whether position is recoverable; it does not trace content back to input pixels.

**Output magnitude.** A strongly attended source can return a short vector. We first measure attention on high-norm tokens: their norm entering the block, before LayerNorm, exceeds three times the image median. One control simply zeros their scores. *Value weighting* instead uses the returned vector at every patch, $c_j = \sum_{h,i} a_{ij}^h \|z_j^h\|_2$, without a threshold. For one query–source pair,

$$\Big\| \sum_h a_{ij}^h z_j^h \Big\|_2 \le \sum_h a_{ij}^h \|z_j^h\|_2.$$

The left side retains cancellation between heads; value weighting sums the right side over queries. These terms describe the attention branch before residuals, MLPs and subsequent token mixing.

## 4 Experiments on three reading assumptions

We compare layers 9–12 of frozen DeiT-S, DeiT-B [29] and DINO ViT-S [8]. Enrichment measures foreground overlap above a same-radius baseline on ImageNet-S (about 12,150 images) [16]; ClickMe compares agreement with own-image and other-image human maps (17,408 images) [14, 21]. Deletion measures how patch removal affects top-1 retention (2,000 images). We specified two comparisons in advance: each change against its matched control, and the combination against received attention (App. A). We require an absolute change of at least 0.02 (alignment) or 0.05 (deletion) and a paired 95% interval excluding zero.

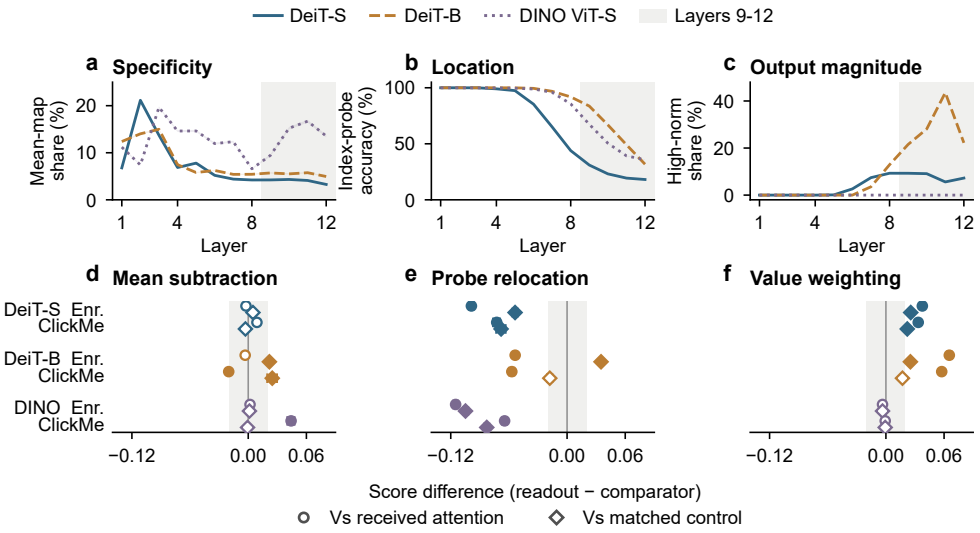



**Figure 2:** Does the phenomenon occur (top), and does changing the readout help (bottom)? Circles compare with received attention; diamonds compare with Gaussian subtraction, matched blur and zeroing high-norm scores, respectively. Bars show paired 95% intervals; filled points meet the decision rule. Grey shading marks layers 9–12 (top) or ±0.02 (bottom). Deletion: Table B.3.

**Is a recurring bright spot informative?** Across layers 6–12, the mean map explained about 3–6% of spatial variance in the DeiTs and up to 17% in DINO (Fig. 2a). In DINO, subtracting the mean raised ClickMe by 0.044, but subtracting a centred Gaussian gave a similar gain. In DeiT-B, mean subtraction beat Gaussian subtraction on all three metrics yet lowered ClickMe by 0.020 relative to received attention. A recurring pattern was not, by itself, a reason to remove it.

**Where should a token's score go?** At layer 12, grid-index accuracy was 18%, 32% and 36% in DeiT-S, DeiT-B and DINO (chance: 0.5%). Yet redistributing scores by decoded positions lowered all nine deep-layer scores. It also did worse than matched blur in DeiT-S and DINO, although it beat blur on DeiT-B enrichment (+0.035). Decodability did not justify relocation.

**What does an active source send?** At layers 9–12, high-norm patches received up to 9% of attention in DeiT-S and 44% in DeiT-B (Fig. 2c). Value weighting improved both alignment scores in both DeiTs. In DeiT-S, enrichment rose by 0.038 and ClickMe rose by 0.033. In DeiT-B the gains were 0.065 in enrichment and 0.058 on ClickMe. Weighting beat zeroing high-norm patches in all but DeiT-B ClickMe. Deletion rose by 0.017 in DeiT-S and 0.029 in DeiT-B: both intervals excluded zero, but neither gain reached 0.05. In DINO, where high-norm patches received no attention in these layers, value weighting changed each score by at most 0.004. For scale, DeiT-S enrichment was 0.178 with value weighting and 0.204 with GAE, which uses class gradients and all 12 layers (Table B.1).

**Do the changes combine?** Combining weighting, relocation and mean subtraction improved only DeiT-B ClickMe, against our prediction of four alignment gains. DeiT-B enrichment was comparable; both DeiT-S scores fell. Value weighting alone helped: did it matter which source received which length?

## 5 Two exploratory tests of what a readout explains

**Which source gets which length?** The placement test asks whether the same lengths would help if assigned to different patches. On 383 Oxford-IIIT Pet test crops with both foreground and background [24], two controls preserve each DeiT-S head's attention and set of lengths at each layer. One borrows the spatial rank order of lengths from another image; the other shuffles lengths only among patches at the same radius. Same-ring enrichment measures foreground alignment at matched radii; AUROC compares foreground with background regardless of radius.

**Table 1:** DeiT-S, layers 9–12, on 383 pet crops. ΔEnrichment is the row minus value weighting (own lengths) (paired 95% intervals).

| Readout | Enrichment | ΔEnrichment [95% CI] | AUROC |
|---|---|---|---|
| Received attention | 0.141 | −0.019 [−0.023, −0.015] | 0.788 |
| Value weighting (own lengths) | 0.160 | — | 0.817 |
| Same lengths in donor rank order | 0.137 | −0.022 [−0.026, −0.019] | 0.787 |
| Same lengths shuffled within radius | 0.138 | −0.021 [−0.025, −0.018] | 0.793 |
| Own lengths, mean attention | 0.120 | −0.040 [−0.048, −0.032] | 0.757 |

Reassigning lengths erased the enrichment gain over received attention (Table 1). Conversely, keeping lengths in place but averaging each head's received attention across patches lowered enrichment by 0.040. Both spatial patterns mattered. The gain could reflect two cues to foreground at the same patches; these comparisons do not isolate a benefit specific to multiplication.

**Which receiving query matters?** The placement tests pooled all queries. If the target is the classifier rather than the foreground, should every query still count? Only the final classification token (CLS) reaches the classifier, so we compared all queries with CLS alone against both targets. On 128 of the pet images, we held attention fixed in block 12, removed each patch's value term from the CLS update in turn, and recomputed the remaining classifier path. For each patch, we measured the absolute change in the logit margin between the original top two classes.

Using only CLS improved intervention ranking in both checkpoints (Table 2), as expected for an intervention on the CLS pathway. The masks tell a different story. In DeiT-S, foreground AUROC fell by 0.091. In DeiT-B, tested on the same images with both outcomes specified in advance, AUROC rose by 0.024 [0.013, 0.036]; the change in same-ring enrichment was uncertain ($+0.005$ $[-0.005, 0.013]$). Matching the query to the classifier did not consistently improve foreground ranking.

**Table 2:** Layer-12 value-weighted rankings, same 128 images. Intervention $\rho$: mean Spearman with CLS margin effects. Differences change only the query set. Full scores and intervals: App. E.3.

| Model | Readout | Intervention $\rho$ | Mask AUROC | Enrichment |
|---|---|---|---|---|
| DeiT-S | Value weighting, all queries | 0.705 | 0.769 | 0.127 |
| | Value weighting, CLS only | 0.747 | 0.678 | 0.095 |
| | **CLS − all queries** | $\mathbf{+0.041}$ | $\mathbf{-0.091}$ | $\mathbf{-0.033}$ |
| DeiT-B | Value weighting, all queries | 0.652 | 0.685 | 0.084 |
| | Value weighting, CLS only | 0.734 | 0.710 | 0.088 |
| | **CLS − all queries** | $\mathbf{+0.082}$ | $\mathbf{+0.024}$ | $\mathbf{+0.005}$ |

## 6 Discussion and conclusion

A bright patch marks a source that received attention. Moving scores to decoded positions lowered all nine model–metric scores. Only value weighting improved both alignment metrics in both DeiTs, and reassigning the lengths erased the enrichment gain on pets. To interpret a map, specify what each patch score measures and which queries it pools, then test the ranking against the outcome the map is meant to explain.

**Limitations and ethical considerations.** Placement was tested in DeiT-S; receiver comparisons cover two checkpoints, one layer and 128 images. An intervention on a contextualised token cannot identify causal input pixels. Object size affects enrichment, and ClickMe may overlap training data.

## References


[1] Samira Abnar and Willem Zuidema. Quantifying Attention Flow in Transformers. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4190–4197, 2020. doi: 10.18653/v1/2020.acl-main.385.

[2] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity Checks for Saliency Maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

[3] Meghna P. Ayyar, Jenny Benois-Pineau, and Akka Zemmari. There is More to Attention: Statistical Filtering Enhances Explanations in Vision Transformers. In *Pattern Recognition. ICPR 2026*, volume 16816 of *Lecture Notes in Computer Science*, pages 591–605. Springer, 2027. doi: 10.1007/978-3-032-31666-0_39.

[4] David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. Network Dissection: Quantifying Interpretability of Deep Visual Representations. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6541–6549, 2017.

[5] Stefan Blücher, Johanna Vielhaben, and Nils Strodthoff. Decoupling Pixel Flipping and Occlusion Strategy for Consistent XAI Benchmarks. *Transactions on Machine Learning Research (TMLR)*, 2024.

[6] Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. Quantizable Transformers: Removing Outliers by Helping Attention Heads Do Nothing. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 75067–75096, 2023. doi: 10.52202/075280-3282.

[7] Gino Brunner, Yang Liu, Damián Pascual, Oliver Richter, Massimiliano Ciaramita, and Roger Wattenhofer. On Identifiability in Transformers. In *International Conference on Learning Representations (ICLR)*, 2020.

[8] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging Properties in Self-Supervised Vision Transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660, 2021.

[9] Hila Chefer, Shir Gur, and Lior Wolf. Transformer Interpretability Beyond Attention Visualization. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 782–791, 2021.

[10] Hila Chefer, Shir Gur, and Lior Wolf. Generic Attention-model Explainability for Interpreting Bi-Modal and Encoder-Decoder Transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 397–406, 2021.

[11] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision Transformers Need Registers. In *International Conference on Learning Representations (ICLR)*, 2024.

[12] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. ImageNet: A Large-Scale Hierarchical Image Database. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.

[13] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[14] Thomas Fel, Ivan Felipe, Drew Linsley, and Thomas Serre. Harmonizing the object recognition strategies of deep neural networks with humans. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022. doi: 10.52202/068431-0685.

[15] Javier Ferrando, Gerard I. Gállego, and Marta R. Costa-jussà. Measuring the Mixing of Contextual Information in the Transformer. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8698–8714, 2022. doi: 10.18653/v1/2022.emnlp-main.595. URL https://aclanthology.org/2022.emnlp-main.595/.

[16] Shanghua Gao, Zhong-Yu Li, Ming-Hsuan Yang, Ming-Ming Cheng, Junwei Han, and Philip Torr. Large-Scale Unsupervised Semantic Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6):7457–7476, 2023. doi: 10.1109/TPAMI.2022.3218275.

[17] Sarthak Jain and Byron C. Wallace. Attention is not Explanation. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3543–3556, 2019. doi: 10.18653/v1/N19-1357.

[18] Nick Jiang, Amil Dravid, Alexei A. Efros, and Yossi Gandelsman. Vision Transformers Don't Need Trained Registers. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025. doi: 10.52202/085713-1891.

[19] Goro Kobayashi, Tatsuki Kuribayashi, Sho Yokoi, and Kentaro Inui. Attention is Not Only a Weight: Analyzing Transformers with Vector Norms. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7057–7075, 2020. doi: 10.18653/v1/2020.emnlp-main.574.

[20] Keying Kuang, Iain Carmichael, and Elizabeth Purdom. When Random Saliency Looks Trained: Architectural Center Bias in CNN Interpretability. In *International Conference on Machine Learning (ICML)*,

volume 306 of *Proceedings of Machine Learning Research*, pages 61236–61266, 2026.

[21] Drew Linsley, Dan Shiebler, Sven Eberhardt, and Thomas Serre. Learning what and where to attend. In *International Conference on Learning Representations (ICLR)*, 2019.

[22] Ali Modarressi, Mohsen Fayyaz, Yadollah Yaghoobzadeh, and Mohammad Taher Pilehvar. GlobEnc: Quantifying Global Token Attribution by Incorporating the Whole Encoder Layer in Transformers. In *Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 258–271, 2022. doi: 10.18653/v1/2022.naacl-main.19. URL https://aclanthology.org/2022.naacl-main.19/.

[23] Hosein Mohebbi, Willem Zuidema, Grzegorz Chrupała, and Afra Alishahi. Quantifying Context Mixing in Transformers. In *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 3378–3400, 2023. doi: 10.18653/v1/2023.eacl-main.245.

[24] Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and Dogs. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3498–3505, 2012. doi: 10.1109/CVPR.2012.6248092.

[25] Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do Vision Transformers See Like Convolutional Neural Networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

[26] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations (ICLR)*, 2021.

[27] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear Transformers Are Secretly Fast Weight Programmers. In *International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 9355–9366, 2021.

[28] Jürgen Schmidhuber. Learning to Control Fast-Weight Memories: An Alternative to Dynamic Recurrent Networks. *Neural Computation*, 4(1):131–139, 1992. doi: 10.1162/neco.1992.4.1.131.

[29] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 10347–10357, 2021.

[30] Matthew Walmer, Saksham Suri, Kamal Gupta, and Abhinav Shrivastava. Teaching Matters: Investigating the Role of Supervision in Vision Transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7486–7496, 2023.

[31] Qishuai Wen, Zhiyuan Huang, Wei He, Xianghan Meng, and Chun-Guang Li. Mixture-of-top-$k$ attention: Efficient attention via scalable fast weights. arXiv:2602.01219v6, 2026.

[32] Sarah Wiegreffe and Yuval Pinter. Attention is not not Explanation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 11–20, 2019. doi: 10.18653/v1/D19-1002.

[33] Ross Wightman. PyTorch Image Models. GitHub repository, https://github.com/huggingface/pytorch-image-models, 2019.

[34] Junyi Wu, Bin Duan, Weitai Kang, Hao Tang, and Yan Yan. Token Transformation Matters: Towards Faithful Post-hoc Explanation for Vision Transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10926–10935, 2024.

[35] Bolei Zhou, Aditya Khosla, Agata Lapedriza, Aude Oliva, and Antonio Torralba. Learning Deep Features for Discriminative Localization. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2921–2929, 2016.

[36] Hongbo Zhu, Theodor Wulff, Rahul Singh Maharjan, Jinpei Han, and Angelo Cangelosi. Noise-Free Explanation for Driving Action Prediction. In *European Conference on Artificial Intelligence (ECAI)*, volume 392 of *Frontiers in Artificial Intelligence and Applications*, pages 451–458, 2024. doi: 10.3233/FAIA240520.

# A Experimental details

**Models.** All models are frozen ViTs with $16\times16$ patches on $224\times224$ inputs. The three baseline backbones come from timm [33]. *DeiT-S and DeiT-B* [29]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224: a crop ratio of 0.9). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224: a crop ratio of 0.875). *DeiT-S with a test-time register* [18]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-$k$ training*: the DeiT-S checkpoint released with MiTA [31] (25 summary slots, $k = 25$), with its own preprocessing (crop ratio 0.875). It differs from our DeiT-S in training and preprocessing, so its row in Table B.1 is not a controlled comparison; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side changes are tested at size S only.

**Attention conventions.** With $x$ the output of the block's first layer norm, projections include the checkpoint's biases:

$$q_i^h = W_Q^h x_i + b_Q^h, \quad k_j^h = W_K^h x_j + b_K^h, \quad v_j^h = W_V^h x_j + b_V^h.$$

Equation 1 includes the output bias $b_O$ and describes the attention branch before residual addition, with dropout inactive at evaluation. $W_O^h$ is the head's slice of the output projection. Value weighting assigns the per-head vector magnitudes to patches; the shared output bias is not assigned to any patch. The fixed-image two-layer representation uses the coupled softmax activation, rather than an elementwise nonlinearity.

**Data.** *Execution test*: 5,000 randomly drawn ImageNet validation images [12]. *ImageNet-S* [16]: all 12,419 validation images with segmentation masks; stored enrichment is finite for 12,150 (12,144 for DINO). All-foreground grids score zero. Restricting enrichment to crops with both classes changed the value-weighting contrast by less than 0.002 in each model; the reported enrichment verdicts use the original eligibility rule. Deletion uses a fixed random subset of 2,000. *ClickMe* [14, 21]: the 17,408-image validation split released with Harmonization (17 `val-*.tfrecords` shards). This split name does not identify the original ImageNet split of each image. Images are resized from $256\times256$ to $224\times224$ without cropping; heatmaps are resized to $64\times64$, blurred with a $10\times10$ Gaussian ($\sigma = 10$), then resized to $224\times224$ (bilinear interpolation). *Held-out set*: 4,000 ImageNet validation images excluded from map-quality and execution evaluation; 2,000 train probes and 2,000 provide mean maps, probe-test accuracies and blur calibration. Our near-duplicate check matched 2 of 4,002 candidate ImageNet validation images to ClickMe records; this does not establish every ClickMe image's training-set membership. Possible overlap with model training images limits generalisation claims.

**Diagnostics.** All three are computed per layer. *Specificity*: with $\bar{s}$ the mean received-attention map of the held-out images and $\mathrm{Var}_j$ the variance over the 196 patches of one image, the share explained is $1 - \sum_x \mathrm{Var}_j(s - \bar{s}) \,/\, \sum_x \mathrm{Var}_j(s)$, summed over the ImageNet-S images $x$. *Location*: the top-1 accuracy of the linear position probe on the 196 patch tokens of each of 2,000 held-out images. *Output magnitude*: the share of received attention on high-norm tokens, on the ImageNet-S images. The high-norm threshold (three times the median, defined below) is an operational definition fixed before the experiments; we did not test other thresholds. The correction itself uses no threshold.

**Corrections and matched controls.** Scores are computed per layer and summed over layers 9–12 without per-layer rescaling, except for calibration of the Gaussian control described below. In the three baseline models, all 197 queries contribute to the readout, which scores only the 196 patch slots. Negative mean-subtracted scores are retained for ranking, without clipping. *Mean-map subtraction* computes $\sum_\ell (s_\ell - \bar{s}_\ell)$ using the 2,000 calibration images. With $S = \sum_\ell s_\ell$ and $\bar{S} = \sum_\ell \bar{s}_\ell$, its control is $S - G\,\mathrm{sd}_j(\bar{S})/\,\mathrm{sd}_j(G)$, where $G$ is a centred isotropic Gaussian ($\sigma = 3.5$ patches). Calibration thus uses the *summed* mean map. *Probe relocation* computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$,

preserving total score. One linear 196-way classifier per model/layer is trained on standardised tokens after the first layer norm: cross-entropy, Adam, learning rate 0.01, 30 epochs, token mini-batches of 4,096, seed 0. Feature means/standard deviations use only the 2,000 training images; standard deviations are floored at $10^{-6}$. Its Gaussian blur control is a row-normalised kernel $K_{jp} \propto \exp(-\|p-j\|^2/(2\sigma_\ell^2))$. The width (0.7–2.9 patches) matches

$$D_\ell = \frac{1}{2000N} \sum_{x,j,p} P_\ell(p \mid x_j)\|p-j\|^2$$

on calibration images, averaging uniformly over source patch indices. ClickMe uses the same crop-trained probe and blur on resized full images, introducing a potential geometry mismatch. *Value weighting* computes $c_j = \sum_{h,i} a_{ij}^h \|W_O^h v_j^h\|_2$, a sum of per-head norm-weighted terms related to Kobayashi et al. [19]. Taking each head's magnitude before summing omits cancellation between heads; residual addition, subsequent normalisation and later-layer computation are outside this score. The control zeros received attention at high-norm patches: the residual-stream block-input norm *before* the first layer norm must exceed three times the image's median patch norm, using the midpoint median. Value weighting itself has no high-norm threshold. *All three corrections*: value weighting, probe relocation, then subtraction of the held-out mean of the resulting map, per layer before summation.

**Metrics.** *Same-ring object enrichment.* A patch is valid if it contains at least one non-ignored pixel, and is an object patch if at least half of its non-ignored pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among all its patches, and average over rings, weighted by their selected patches. Invalid patches are excluded from selection and ring baselines. Ties are broken with the fixed seeded perturbation used in the original runs. A purely radial map scores zero in expectation over tie-breaking. *Image-specific ClickMe alignment.* The map is upsampled to $224\times224$ and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the previous image's human map within each batch (`roll(1)`; batch size 64). For an image-independent map, these differences average to zero over the paired set because the control permutes the same human maps; individual-image differences need not be zero. This removes fixed-prior alignment on average, without guaranteeing invariance to a prior mixed into an image-dependent map. *Deletion score.* Patches are ranked once by the original map. At removal budgets $0, 1, 2, 4, 8, 16, 32, 64, 98, 147$, we record the binary indicator that the original top-1 prediction is preserved. Tokens are dropped after positional embedding; CLS and any register remain. We use an SRG-style LIF–MIF area difference [5], adapting the value function from class probability to top-1 preservation. Areas use trapezoidal integration over removed-patch fraction $b/196$, from 0 to $147/196$. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps have exactly zero mean on the latter, up to numerical precision); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired image-bootstrap with 1,000 resamples, seed 0 and 95% percentile intervals. A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its absolute mean reaches 0.02 (0.05 for deletion), and *comparable* (*c*) otherwise. The comparable category also covers imprecise differences. Intervals are unadjusted for multiple comparisons. Resampling holds checkpoints, probes and the within-batch ClickMe pairing fixed, excluding uncertainty from training, probe fitting or alternative pairings. Because the control reuses human maps within each batch, ClickMe score rows are not independent pairings. For the ClickMe sensitivity check, resampling the 272 original batches preserves within-batch dependence. Against zeroing high-norm patches, value weighting gives intervals of [0.021, 0.023] in DeiT-S and [0.016, 0.018] in DeiT-B.

**GAE reference.** The gradient-based reference [10] targets the unmodified model's predicted top-1 logit and uses all 12 attention layers. Starting with $R = I$, each layer updates $R \leftarrow R + \mathrm{mean}_h[(\nabla_{A^h} f_c \odot A^h)_+]R$, then takes the final CLS-to-patch row. It is class-specific and spans all layers, whereas the four readouts are class-agnostic and sum layers 9–12. Its scores provide descriptive context rather than a matched single-component ablation.

**Registered predictions.** Table A.1 lists predictions with registered verdicts. We fixed the protocol after smoke tests on 32–64 images and before the formal runs; earlier observations are disclosed in

the registration. P1 and P2 replicate earlier observations. P3 predicted four alignment gains from combining the changes. Four higher scores meant "holds", one to three "partly holds", and none "does not hold". One higher, one comparable and two lower scores thus count as "partly holds" but do not support a general improvement. P5 describes last-layer recovery. P7 specifies primary comparisons: the combination against received attention and each change against its control, with no predicted direction for the latter. Other individual comparisons against received attention are descriptive except for P4a and P4b. Supplementary mechanism analyses are post hoc.

**Table A.1:** Registered predictions and outcomes (centre-controlled metrics: enrichment and ClickMe alignment).

| | Prediction | Outcome |
|---|---|---|
| P4a | Value weighting is comparable to received attention in DINO ViT-S on all three metrics | holds (3 of 3 cells) |
| P4b | Relocation at layers 1–4 is comparable to received attention on the centre-controlled metrics in all models | holds (6 of 6 cells) |
| P3 | All three changes together are higher than received attention on the centre-controlled metrics in DeiT-S and DeiT-B | partly holds (1 higher: DeiT-B ClickMe; 1 comparable; 2 lower) |
| P6 | Neither model-side change is higher than DeiT-S received attention on the centre-controlled metrics | holds (4 of 4 cells) |
| P1 | Execution agreement rises with $k$ and exceeds that of random sets at every $k$ | holds in all models |
| P2 | Probe accuracy falls with depth; the share of attention on high-norm tokens peaks in layers 6–12 of the supervised models and is below 1% in every layer of DINO ViT-S | holds in all models |

**Why test a register and MiTA checkpoint?** These are model-side alternatives to changing how a frozen backbone is read. A test-time register moves selected feed-forward activations to an added token, reducing high-norm patch artefacts. The released MiTA checkpoint changes attention and training. Neither is an ablation of the two-layer representation, which also describes our unchanged baseline. P6 predicted that neither alternative would exceed the alignment margin at size S; all four comparisons were not higher. MiTA's different training and preprocessing restrict interpretation of that comparison.

**Compute and code.** The original runs used approximately 19 NVIDIA L4 GPU hours. The code repository links each display to its data and implementation and gives separate instructions for rebuilding from saved scores and rerunning inference.

## B Additional results

We first report absolute scores and execution checks, then the paired differences used in Fig. 2. Additional analyses test whether map changes track high-norm attention and whether the maps depend on trained parameters.

**Table B.1:** Complete scores for the original tests, layers 9–12 unless indicated. Arrows mark higher/lower scores versus received attention (DeiT-S for model-side rows). Diamonds indicate higher scores than the matched control. Bold indicates higher than both. Untagged tested cells are comparable. GAE uses gradients and all 12 layers. Dashes indicate unavailable comparisons.

| | DeiT-S | | | DeiT-B | | | DINO ViT-S | | |
|---|---|---|---|---|---|---|---|---|---|
| Reading | Enr. | Click. | Del. | Enr. | Click. | Del. | Enr. | Click. | Del. |
| Received attention | 0.140 | 0.157 | 0.236 | 0.080 | 0.107 | 0.169 | 0.199 | 0.208 | 0.285 |
| Value weighting | **0.178**$^{\uparrow\diamond}$ | **0.190**$^{\uparrow\diamond}$ | 0.253 | **0.146**$^{\uparrow\diamond}$ | 0.165$^{\uparrow}$ | 0.198 | 0.196 | 0.207 | 0.285 |
| zero high-norm scores | 0.152 | 0.168 | 0.241 | 0.121$^{\uparrow}$ | 0.148$^{\uparrow}$ | 0.186 | 0.199 | 0.208 | 0.285 |
| Mean-map subtraction | 0.138 | 0.165 | 0.208 | 0.077$^{\diamond}$ | 0.087$^{\downarrow\diamond}$ | 0.144$^{\diamond}$ | 0.201 | 0.252$^{\uparrow}$ | 0.233$^{\downarrow}$ |
| subtract centre Gaussian | 0.133 | 0.168 | 0.167$^{\downarrow}$ | 0.056$^{\downarrow}$ | 0.062$^{\downarrow}$ | 0.024$^{\downarrow}$ | 0.200 | 0.253$^{\uparrow}$ | 0.232$^{\downarrow}$ |
| Probe relocation | 0.041$^{\downarrow}$ | 0.084$^{\downarrow}$ | 0.076$^{\downarrow}$ | 0.027$^{\downarrow\diamond}$ | 0.050$^{\downarrow}$ | 0.051$^{\downarrow}$ | 0.084$^{\downarrow}$ | 0.143$^{\downarrow}$ | 0.141$^{\downarrow}$ |
| matched blur | 0.095$^{\downarrow}$ | 0.152 | 0.181$^{\downarrow}$ | −0.008$^{\downarrow}$ | 0.068$^{\downarrow}$ | 0.097$^{\downarrow}$ | 0.189 | 0.226 | 0.287 |
| All three corrections | 0.059$^{\downarrow}$ | 0.123$^{\downarrow}$ | 0.076$^{\downarrow}$ | 0.079 | 0.137$^{\uparrow}$ | 0.085$^{\downarrow}$ | 0.088$^{\downarrow}$ | 0.177$^{\downarrow}$ | 0.090$^{\downarrow}$ |
| + test-time register | 0.152 | 0.169 | 0.236 | – | – | – | – | – | – |
| Top-$k$ training (MiTA) | 0.100$^{\downarrow}$ | 0.115$^{\downarrow}$ | – | – | – | – | – | – | – |
| GAE (reference) | 0.204 | 0.234 | 0.345 | 0.196 | 0.213 | 0.308 | 0.221 | 0.234 | 0.418 |

**Execution test.** The $14\times14$ grid is pooled into 16 overlapping regions. In every layer/head, each region's mean query selects its top-$k$ keys; CLS retains its own set. Sets from an unmodified forward pass restrict every query to its region's set in all 12 layers simultaneously. Top-$k$ sets beat equal-size random sets at every tested $k$ (4–128) in all models, and agreement rises with $k$ (Fig. B.1).

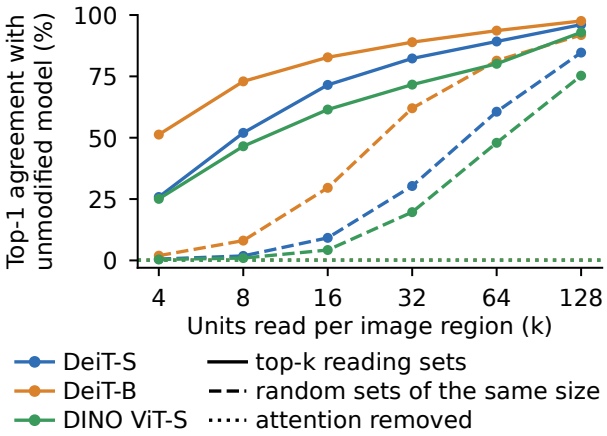



**Figure B.1:** Top-1 agreement on 5,000 ImageNet images with region-selected top-$k$ units (solid), random units (dashed) or attention removed (dotted). Stored 95% intervals are at most $\pm1.5$ percentage points. Dotted curves use crop ratio 0.875 for all models; other curves use each checkpoint's main preprocessing.

**Diagnostics per layer.** Distance from the centre explained 82–98% of mean-map variance at layers 9–12. Before registration, we also measured raw value lengths on 11,919 ImageNet-S images (crop ratio 0.875; main DeiT runs: 0.9). Within each eligible image and head we divided the mean raw value length of high-norm patches by the median for ordinary patches. Averaging first within layers, then equally over layers 9–12, gave 0.29 for DeiT-S and 0.23 for DeiT-B. Only images containing high-norm patches contribute. The block-input norm defines those patches before LayerNorm; raw value length is measured after LayerNorm and value projection.

**Recovery at the last layer.** In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (ClickMe values in Table B.2). Applied to layer 12, value weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

**Table B.2:** Per-layer diagnostics (Fig. 2a–c: ImageNet-S for the mean-map and high-norm shares; 2,000 held-out images for probe accuracy) and uncorrected ClickMe alignment (last three columns). S, B, D: DeiT-S, DeiT-B, DINO ViT-S; probe chance: 0.5%. Values are rounded; DINO layers 9–12 have exactly zero high-norm attention on this dataset.

| | Specificity: mean-map share (%) | | | Location: probe accuracy (%) | | | Output magnitude: high-norm share (%) | | | Map quality: ClickMe alignment | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Layer | S | B | D | S | B | D | S | B | D | S | B | D |
| 1 | 6.8 | 12.4 | 11.2 | 100.0 | 100.0 | 99.9 | 0.0 | 0.0 | 0.0 | −0.108 | −0.111 | −0.089 |
| 2 | 21.2 | 14.0 | 7.4 | 100.0 | 100.0 | 100.0 | 0.0 | 0.0 | 0.0 | 0.007 | 0.041 | 0.064 |
| 3 | 13.7 | 15.0 | 19.7 | 99.9 | 100.0 | 99.9 | 0.0 | 0.0 | 0.0 | 0.081 | 0.004 | 0.056 |
| 4 | 6.9 | 7.5 | 14.6 | 99.0 | 100.0 | 99.8 | 0.0 | 0.0 | 0.0 | 0.132 | 0.110 | 0.140 |
| 5 | 7.8 | 5.8 | 14.6 | 97.5 | 99.9 | 99.6 | 0.1 | 0.0 | 0.0 | 0.148 | 0.130 | 0.150 |
| 6 | 5.2 | 6.2 | 11.9 | 85.3 | 99.5 | 98.9 | 2.6 | 0.0 | 0.0 | 0.136 | 0.151 | 0.167 |
| 7 | 4.4 | 5.4 | 12.4 | 65.0 | 96.8 | 95.9 | 7.4 | 3.5 | 0.0 | 0.147 | 0.131 | 0.150 |
| 8 | 4.2 | 5.4 | 6.6 | 43.9 | 91.8 | 85.6 | 9.3 | 12.9 | 0.0 | 0.149 | 0.118 | 0.170 |
| 9 | 4.2 | 5.7 | 9.5 | 31.0 | 83.6 | 67.4 | 9.3 | 21.5 | 0.0 | 0.184 | 0.112 | 0.198 |
| 10 | 4.3 | 5.5 | 15.2 | 23.1 | 66.8 | 50.5 | 9.1 | 28.1 | 0.0 | 0.171 | 0.139 | 0.207 |
| 11 | 4.1 | 5.8 | 16.7 | 19.4 | 49.2 | 39.2 | 5.6 | 43.5 | 0.0 | 0.149 | 0.120 | 0.211 |
| 12 | 3.3 | 5.0 | 13.6 | 18.1 | 31.8 | 35.7 | 7.2 | 22.2 | 0.0 | 0.098 | 0.023 | 0.179 |

**Table B.3:** Paired differences, layers 9–12: mean, 95% image-bootstrap interval and verdict (*h*/*c*/*l*: higher/comparable/lower). Margins: 0.02 (enrichment/ClickMe), 0.05 (deletion); intervals unadjusted. Samples: App. A. Top: each change (descriptive) and their combination (primary) versus received attention. Bottom: each change versus its matched control (primary). Registered negative controls are also discussed in App. A.

| Model | Same-ring enrichment | ClickMe alignment | Deletion score |
|---|---|---|---|
| *Value weighting − received attention* | | | |
| DeiT-S | 0.038 [0.036, 0.039] *h* | 0.033 [0.032, 0.034] *h* | 0.017 [0.013, 0.021] *c* |
| DeiT-B | 0.065 [0.063, 0.067] *h* | 0.058 [0.056, 0.059] *h* | 0.029 [0.024, 0.034] *c* |
| DINO ViT-S | −0.004 [−0.004, −0.003] *c* | −0.001 [−0.001, −0.001] *c* | 0.000 [−0.003, 0.004] *c* |
| *Mean-map subtraction − received attention* | | | |
| DeiT-S | −0.002 [−0.003, −0.002] *c* | 0.009 [0.008, 0.010] *c* | −0.028 [−0.034, −0.023] *c* |
| DeiT-B | −0.003 [−0.004, −0.003] *c* | −0.020 [−0.021, −0.019] *l* | −0.025 [−0.031, −0.019] *c* |
| DINO ViT-S | 0.002 [0.001, 0.003] *c* | 0.044 [0.040, 0.048] *h* | −0.052 [−0.059, −0.044] *l* |
| *Probe relocation − received attention* | | | |
| DeiT-S | −0.099 [−0.101, −0.097] *l* | −0.073 [−0.076, −0.069] *l* | −0.161 [−0.171, −0.150] *l* |
| DeiT-B | −0.054 [−0.055, −0.052] *l* | −0.057 [−0.059, −0.055] *l* | −0.118 [−0.127, −0.110] *l* |
| DINO ViT-S | −0.115 [−0.117, −0.113] *l* | −0.064 [−0.067, −0.062] *l* | −0.144 [−0.153, −0.134] *l* |
| *All three corrections − received attention* | | | |
| DeiT-S | −0.081 [−0.083, −0.079] *l* | −0.034 [−0.038, −0.030] *l* | −0.160 [−0.171, −0.149] *l* |
| DeiT-B | −0.001 [−0.003, 0.001] *c* | 0.030 [0.027, 0.032] *h* | −0.084 [−0.093, −0.076] *l* |
| DINO ViT-S | −0.111 [−0.114, −0.109] *l* | −0.031 [−0.035, −0.027] *l* | −0.195 [−0.205, −0.184] *l* |
| *Value weighting − zero high-norm scores (its control)* | | | |
| DeiT-S | 0.026 [0.025, 0.027] *h* | 0.022 [0.021, 0.023] *h* | 0.012 [0.008, 0.016] *c* |
| DeiT-B | 0.025 [0.024, 0.026] *h* | 0.017 [0.016, 0.018] *c* | 0.013 [0.008, 0.018] *c* |
| DINO ViT-S | −0.004 [−0.004, −0.003] *c* | −0.001 [−0.001, −0.001] *c* | 0.000 [−0.003, 0.004] *c* |
| *Mean-map subtraction − centre-Gaussian subtraction (its control)* | | | |
| DeiT-S | 0.005 [0.004, 0.005] *c* | −0.003 [−0.005, −0.001] *c* | 0.040 [0.034, 0.047] *c* |
| DeiT-B | 0.022 [0.021, 0.023] *h* | 0.025 [0.020, 0.030] *h* | 0.120 [0.109, 0.130] *h* |
| DINO ViT-S | 0.001 [0.001, 0.002] *c* | −0.001 [−0.001, 0.000] *c* | 0.001 [−0.003, 0.005] *c* |
| *Probe relocation − matched blur (its control)* | | | |
| DeiT-S | −0.054 [−0.056, −0.051] *l* | −0.068 [−0.073, −0.064] *l* | −0.105 [−0.116, −0.095] *l* |
| DeiT-B | 0.035 [0.034, 0.036] *h* | −0.018 [−0.020, −0.015] *c* | −0.047 [−0.055, −0.037] *c* |
| DINO ViT-S | −0.105 [−0.107, −0.102] *l* | −0.083 [−0.085, −0.080] *l* | −0.146 [−0.155, −0.137] *l* |

**Differences behind Table B.1.** Table B.3 gives the paired differences with their intervals. The values in Table B.1 have 95% intervals of at most $\pm 0.005$ on enrichment and ClickMe alignment and $\pm 0.012$ on deletion. The model-side differences from DeiT-S received attention are $+0.012$ [0.011, 0.013] on enrichment, $+0.013$ [0.012, 0.013] on ClickMe and 0.000 [$-0.005$, 0.004] on deletion for the test-time register, and $-0.040$ [$-0.042$, $-0.039$] and $-0.042$ [$-0.044$, $-0.040$] on the first two for top-$k$ training. The paired MiTA enrichment contrast uses 12,118 images valid for both models.

**What drives the value-weighting gain?** Value weighting reduces the relative weight of attention attached to small transformed-value vectors. One consequence was registered (P4a); additional checks were post hoc. *No practically meaningful gain in DINO (P4a).* In DINO ViT-S, with zero high-norm attention at layers 9–12 on ImageNet-S, every difference from received attention is at most 0.004 in size (Table B.3). *High-norm attention and alignment gain (post hoc).* Across images, gains have a weak positive association with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *Assigned versus shuffled weights (post hoc).* Against the same weights shuffled among the patches of each head, value weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. The shuffle control averages seeds 0–4 within each image before bootstrap resampling. Projected minus raw value weighting ranged from $-0.0104$ to $-0.0004$ across the six model–alignment comparisons. None reached the 0.02 margin; the study did not test equivalence. Value weighting aggregates per-head terms related to Kobayashi et al. [19] and shares a norm-weighted component with Zhu et al. [36]; their full analyses and aggregation differ.

### B.1 Parameter-randomisation sanity check

The original study included a model-parameter randomisation check [2] on 1,000 fixed-seed ImageNet-S images. Parameters were cumulatively re-initialised from the classifier and final norm through blocks 12–1 to the patch/position embeddings and CLS token. The trained model's probes and mean maps remained fixed. Received attention and its corrections are class-agnostic and therefore do not depend on classifier parameters alone. Table B.4 reports the stored median per-image Spearman correlations between trained and fully randomised maps, with layers 9–12 summed. All five readings meet the registered median $< 0.5$ criterion; the model-independent centre Gaussian fails as designed. This is a necessary sensitivity check, not proof of prediction faithfulness.

**Table B.4:** Stored model-randomisation results: median map correlation after full randomisation. The five readings pass the registered $< 0.5$ criterion in every model.

| Reading | DeiT-S | DeiT-B | DINO ViT-S |
|---|---|---|---|
| Received attention | $-0.033$ | 0.055 | 0.071 |
| Mean-map subtraction | 0.187 | 0.366 | 0.100 |
| Probe relocation | 0.063 | 0.004 | $-0.002$ |
| Value weighting | $-0.064$ | 0.070 | 0.114 |
| All three corrections | 0.069 | 0.078 | 0.025 |
| Centre Gaussian | 1.000 | 1.000 | 1.000 |

### B.2 Alternative deletion endpoints

**Deletion endpoints.** For each image and method, trapezoidal integration uses the original removal budgets, divided by 196. The probability endpoint integrates the probability of the original top-1 class under least-important-first removal minus most-important-first removal. The KL endpoint uses the reverse difference in divergence from the original prediction, so that larger scores again indicate greater separation. For both endpoints the paired curve difference is zero at budget zero. These measures have no registered practical margin. An advantage in integrated area can reverse at a small removal budget; see the budget-wise curves and intervals.

**Table B.5:** Exploratory endpoint differences (value weighting minus comparator). Probability and KL columns give the mean paired area and 95% image-bootstrap interval over the fixed 2,000-image deletion subset. ClickMe uses the original 17,408 images with batch-bootstrap intervals. KL has units of nats times fraction removed.

| Model | Comparator | Probability area | KL area | ClickMe |
|---|---|---|---|---|
| DeiT-S | Received attention | 0.014 [0.012, 0.015] | 0.058 [0.050, 0.066] | 0.033 [0.032, 0.034] |
| DeiT-S | Zero high-norm | 0.010 [0.009, 0.012] | 0.041 [0.033, 0.049] | 0.022 [0.021, 0.023] |
| DeiT-B | Received attention | 0.022 [0.020, 0.024] | 0.107 [0.096, 0.118] | 0.058 [0.056, 0.059] |
| DeiT-B | Zero high-norm | 0.010 [0.008, 0.012] | 0.048 [0.038, 0.058] | 0.017 [0.016, 0.018] |
| DINO | Received attention | 0.000 [−0.001, 0.002] | −0.003 [−0.009, 0.003] | −0.001 [−0.001, −0.001] |
| DINO | Zero high-norm | 0.000 [−0.001, 0.002] | −0.003 [−0.009, 0.003] | −0.001 [−0.001, −0.001] |

### B.3 Object geometry and the enrichment scale

For an image with $M$ valid patches and $k > 0$ foreground patches, let ring $r$ contain $m_r$ valid patches of which $f_r$ are foreground. Selecting a patch in this ring contributes $(y - f_r/m_r)/k$ to enrichment, where $y$ is its foreground indicator. Foreground contributions are nonnegative and background contributions are nonpositive. Selecting all $k$ foreground patches therefore attains the maximum

$$E_{\max} = 1 - \frac{1}{k} \sum_{r:m_r>0} \frac{f_r^2}{m_r} \;\le\; 1 - \frac{k}{M}.$$

The inequality follows by Cauchy–Schwarz. It bounds the score itself. The gain over received attention also depends on its starting score. A large raw improvement for small objects can thus reflect properties of both the map and the metric.

**Area and centroid displacement.** Object area is the share of valid patch slots labelled foreground. Centroid displacement is the distance from their centroid to the grid centre, in patches. For each checkpoint we computed the empirical quartiles on its finite-geometry images, assigning ties to the lower bin. The DeiT area cut points were 0.158, 0.342 and 0.571. DINO's different crop produced different masks. Intervals use 2,000 paired image resamples from the same supplementary run. In DeiT-S and DeiT-B the largest-minus-smallest area quartile differences in the value weighting gain were $-0.090$ and $-0.157$ for enrichment. Comparing the most off-centre with the most central quartile instead gave $+0.057$ and $+0.095$. Area and centroid displacement covary. Among the smallest objects the latter differences were $+0.041$ [0.028, 0.053] and $+0.047$ [0.028, 0.066]. Some joint bins are small and these are observational contrasts.

**Metric sensitivity.** After seeing the enrichment gradients we examined patch AUROC under the same quartile cut points. AUROC is the probability that a foreground patch outranks a background patch, counting ties as one half. In the supervised models the gain in AUROC persists across area quartiles (Table B.6). AUROC does not condition on radius, so this comparison changes spatial adjustment as well as how foreground patches are selected. Relative to received attention, value weighting changes AUROC by $+0.028$, $+0.046$ and $-0.005$ for DeiT-S, DeiT-B and DINO.

**Table B.6:** Mean value weighting minus received attention by foreground-area quartile on ImageNet-S. Q1 contains the smallest objects. Both metrics use the same within-model cut points. AUROC excludes all-foreground crops.

| Metric | Model | Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|---|---|
| Enrichment | DeiT-S | 0.092 | 0.041 | 0.013 | 0.002 |
| Enrichment | DeiT-B | 0.165 | 0.060 | 0.026 | 0.008 |
| Enrichment | DINO | −0.003 | −0.004 | −0.005 | −0.003 |
| AUROC | DeiT-S | 0.024 | 0.030 | 0.031 | 0.026 |
| AUROC | DeiT-B | 0.042 | 0.048 | 0.048 | 0.046 |
| AUROC | DINO | 0.000 | −0.002 | −0.006 | −0.013 |

## C Tests of spatial relocation

Relocation lowered all nine deep-layer scores relative to received attention (Table B.1). Two planned comparisons narrow the problem. Early relocation remains comparable under the registered rule (P4b): at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. In DeiT-S and DINO ViT-S, it also scored below blur matched for mean squared displacement (Table B.3). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none meets its specified criterion. Only 2–29% of the probe's error mass lay more than three patches from the true slot. Replacing the linear probe with the tested MLP did not restore alignment. The probe targets index decodability rather than input-content provenance; whether this mismatch explains the failed relocation remains untested (Section 3).

**Table C.1:** Four explanations for probe-relocation failure (post hoc). Model order: DeiT-S, DeiT-B, DINO ViT-S. Cells are the two centre-controlled metrics in the three models; criteria and probe settings are defined in this section.

| Explanation | Test (layers 9–12 unless stated) | Result |
|---|---|---|
| The probe sends score to distant patches | Share of the probe's error mass more than three patches from the true patch | Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies in the $3\times3$ neighbourhood of the true patch |
| The probe has a fixed position preference | Peak of the probe's mean prediction relative to uniform; its correlation with the mean map | Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map |
| The linear probe is too weak | Two-layer MLP probe in place of the linear one | Layer-12 accuracy rises by only 1.8, 3.8 and 6.7 points; relocation is still lower than received attention in all six cells |
| Global tokens are harder to place | Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12) | No consistent relation; accuracy falls with globality only in DeiT-B |

**Probe capacity and explanatory criteria.** The tested MLP has a 1,024-unit hidden layer, GELU and a 196-way output. It uses the linear probe's optimiser, epochs, minibatches and standardisation, trained on 1,800 of the original probe-training images with 200 for best-validation-epoch selection; the original 2,000 test images remain separate. After observing the original relocation result, we specified the following criteria before running these follow-up tests: error mass beyond three patches greater than 0.5 at every deep model/layer; fixed preference peak at least twice uniform and mean-map correlation at least 0.5 (with positive lower interval bound) in every deep model/layer; MLP layer-12 accuracy gain above 10 percentage points in some model, with relocation rescued if it is not lower in at least 4/6 alignment comparisons. None of these criteria was met. This MLP comparison leaves other probe designs and calibration errors open.

**Globality analysis.** Excluding high-norm tokens, globality is a token's cosine similarity to the mean of the other retained tokens in the image. Tokens are split into five equal-size rank groups. We relate the group rank to index accuracy and probe entropy, overall and within four equal-count radial groups, at layers 10–12. After observing the original relocation result, we set a follow-up criterion requiring negative accuracy and positive entropy associations (overall intervals and within-ring means) in at least two models. No consistent relationship meets this criterion.

## D Example maps

We drew 24 ClickMe images at random with a fixed seed. Fig. D.1 shows the two with the smallest ClickMe index, and Fig. D.2 the next four. Received attention and value weighting look much alike, while probe relocation spreads the score over many more patches. Because each panel has its own colour scale, use Table B.1 to assess changes in patch ranking across images.

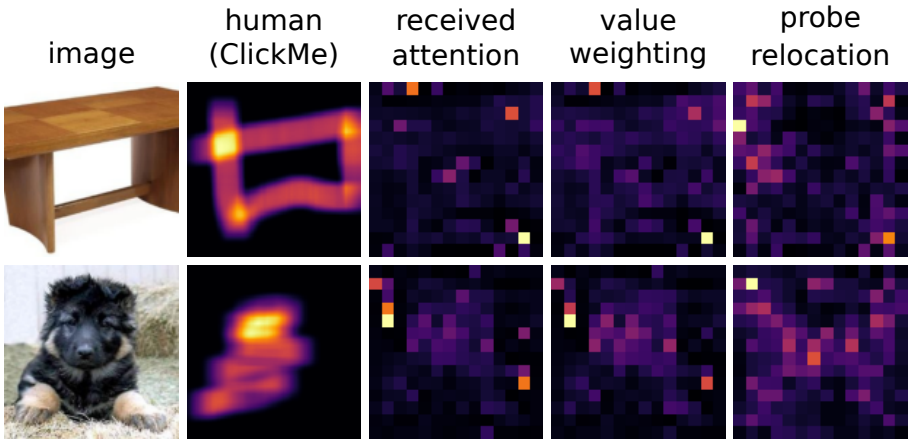



**Figure D.1:** The first two fixed-seed examples, DeiT-S, layers 9–12. Colours show scores min–max normalised within each map; amplitudes are not comparable across panels. These images were not selected for correction gain.

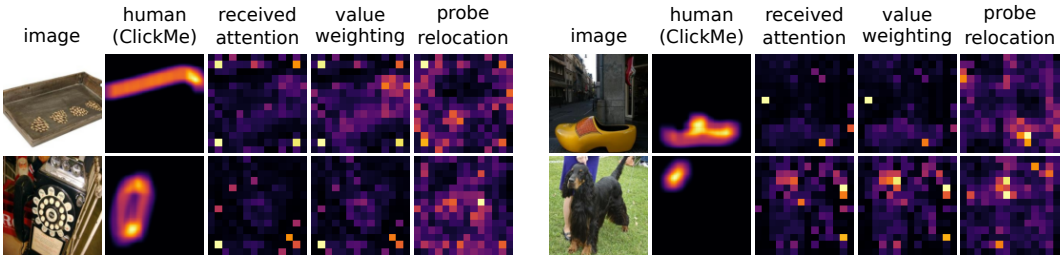



**Figure D.2:** DeiT-S on four further ClickMe images, layers 9–12; colours are normalised within each map and do not compare amplitudes across panels.

# E Details of the two exploratory tests

## E.1 Independent pet-image controls

We selected the first 384 Oxford test images in fixed filename-hash order. We resized the short side to 248 pixels and centre-cropped to 224, using bicubic interpolation for images and nearest-neighbour for trimaps. We ignored boundary pixels and excluded patches with no valid pixels. A patch was foreground if at least half its valid pixels were foreground. One crop contained only foreground. Adjacent images in hash order exchanged length ranks, preserving each recipient's lengths within each layer and head. Within-ring shuffles used the same permutation across heads for a given image and layer. The reverse control keeps projected lengths at their own patches. Within each image, layer and head it replaces received attention at every patch with the same mean over the 196 patch slots. This preserves each head's total patch attention but removes its spatial variation.

We evaluated all seven readouts on the same 383 eligible crops. Table 1 reports paired differences in the primary endpoint, same-ring enrichment; the table below gives corresponding differences in AUROC. Both use 95% percentile intervals from 10,000 paired image-bootstrap draws (seed 2026), without a practical margin. A follow-up sensitivity check resampled the 192 donor pairs, weighting by the number of eligible images in each draw. Value weighting minus donor-rank gives 0.0223 [0.0187, 0.0261] in this check.

**Table E.1:** All seven spatial readouts on the same 383 pet crops. Each AUROC difference subtracts the AUROC of value weighting at the original positions. Mean attention preserves each head's total patch attention; net output sums head vectors before taking each query–source magnitude, then pools queries.

| Reading | AUROC | ΔAUROC [95% CI] | Enrichment |
|---|---|---|---|
| Received attention | 0.788 | −0.028 [−0.031, −0.026] | 0.141 |
| Value weighting (own lengths) | 0.817 | — | 0.160 |
| Same lengths in donor rank order | 0.787 | −0.029 [−0.032, −0.026] | 0.137 |
| Same lengths shuffled within radius | 0.793 | −0.023 [−0.025, −0.021] | 0.138 |
| Own lengths, mean attention | 0.757 | −0.059 [−0.069, −0.049] | 0.120 |
| Unprojected value norm | 0.822 | 0.006 [0.005, 0.006] | 0.163 |
| Norm after summing heads | 0.809 | −0.008 [−0.009, −0.007] | 0.156 |

## E.2 Last-block CLS value intervention

A second fixed filename hash selected 128 of the 384 images independently of their scores. At block 12 we cached its input and the unmodified logits. For each patch $j$ we held the original CLS attention weights fixed and removed $t_j = \sum_h a^h_{\mathrm{CLS},j} z^h_j$ from the CLS attention output. The shared projection bias stayed in place. The remaining residual, token-wise MLP, final normalisation and classifier were evaluated for each modified CLS. If $m$ is the margin between the original highest and second-highest logits, the response was $|m_{-j} - m|$ for those same classes. This intervention concerns the contextualised token at that slot. Its input evidence can span multiple image regions.

We ranked the 196 patch responses within each image and compared each readout with that ranking using Spearman correlation. Independent full forward passes on four images at three patches each matched the cached logits to within $2.4 \times 10^{-6}$. Table 2 reports the paired query comparison.

**Matched mask comparison.** After inspecting the classifier intervention we evaluated the saved layer-12 rankings on the same 128 images and masks. All 128 had eligible foreground and background patches. Table 2 reports the three outcomes with equal image weights. Keeping the CLS query fixed, moving from the sum of per-head magnitudes to net magnitude changed margin Spearman by $+0.0092$ [0.0069, 0.0116], enrichment by $+0.0037$ [0.0007, 0.0069] and AUROC by $+0.0002$ [$-0.0031$, 0.0033].

## E.3 Receiver comparison in DeiT-B

In DeiT-S, restricting the readout to CLS improved intervention ranking but reduced foreground AUROC. We tested whether both directions persisted in a second checkpoint. Before scoring DeiT-B, we specified both endpoints in `experiments/receiver_review/`. We retained the same 128 pet images, preprocessing, block 12, five readouts, 196 interventions per image and bootstrap settings.

All 128 crops contained both mask classes. This follow-up was exploratory relative to the main registration.

Independent full forward passes on four images at patches 0, 97 and 195 matched the cached logits and class margins to within $3 \times 10^{-6}$ (tolerance: 0.002).

**Table E.2:** All five layer-12 rankings on the same 128 images. Intervention $\rho$ measures agreement with CLS-path sensitivity; AUROC and enrichment compare the unmodified score maps with foreground masks.

| Model | Readout | Intervention $\rho$ | Mask AUROC | Enrichment |
|---|---|---|---|---|
| DeiT-S | All queries, attention | 0.696 | 0.754 | 0.117 |
| DeiT-S | Value weighting, all queries | 0.705 | 0.769 | 0.127 |
| DeiT-S | Value weighting, CLS only | 0.747 | 0.678 | 0.095 |
| DeiT-S | CLS, net vector | 0.756 | 0.678 | 0.098 |
| DeiT-S | CLS, attention | 0.730 | 0.666 | 0.081 |
| DeiT-B | All queries, attention | 0.589 | 0.649 | 0.051 |
| DeiT-B | Value weighting, all queries | 0.652 | 0.685 | 0.084 |
| DeiT-B | Value weighting, CLS only | 0.734 | 0.710 | 0.088 |
| DeiT-B | CLS, net vector | 0.751 | 0.719 | 0.096 |
| DeiT-B | CLS, attention | 0.675 | 0.676 | 0.062 |

**Table E.3:** Paired effects of retaining only the CLS query in the sum of head magnitudes. Intervals are 95% percentile intervals from 10,000 image-bootstrap draws, seed 2026; no registered practical margin is applied.

| Model | Endpoint | CLS − all queries [95% CI] |
|---|---|---|
| DeiT-S | Intervention $\rho$ | 0.041 [0.026, 0.057] |
| DeiT-S | Mask AUROC | −0.091 [−0.110, −0.074] |
| DeiT-S | Enrichment | −0.033 [−0.043, −0.023] |
| DeiT-B | Intervention $\rho$ | 0.082 [0.068, 0.096] |
| DeiT-B | Mask AUROC | 0.024 [0.013, 0.036] |
| DeiT-B | Enrichment | 0.005 [−0.005, 0.013] |

Retaining cancellation between heads is another distinction: in DeiT-B, net magnitude versus summed head magnitudes changes intervention correlation by 0.017 [0.014, 0.020], AUROC by 0.009 [0.007, 0.011] and enrichment by 0.008 [0.005, 0.011].

# F Key parts of the code

Given the attention $a_{ij}^h$ and values $v_j^h$ of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,T,T]; v: [B,H,T,d]; w_o: [D,H,d]
    # T includes CLS; values include the value-projection bias.
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v)
    length = length.clamp_min(0).sqrt()
    received = p.sum(2)  # sum over queries
    s = received.sum(1)[:, prefix:]
    c = (received * length).sum(1)[:, prefix:]
    return s, c

def relocate(s, probs):  # source j to predicted position p
    return torch.einsum('bj,bjp->bp', s, probs)
```

For the independent pet experiment, the following readouts vary the placement of lengths or remove spatial variation in received attention. Here $a$ has axes (image, head, query, patch), and $z$ contains each head's projected value vector. The implementation processes queries in chunks.

```
# Own lengths versus the same lengths in a donor image's spatial order
length = z.norm(dim=-1)                          # [B, H, 196]
order = donor_length.argsort(dim=-1)
borrowed = torch.empty_like(length).scatter(-1, order, length.sort(-1).values)
received = a.sum(dim=2)
own_map = (received * length).sum(dim=1)
donor_map = (received * borrowed).sum(dim=1)
mean_attention_map = (received.mean(dim=-1, keepdim=True) * length).sum(dim=1)

# Magnitude after summing the head vectors for each query and patch
net_map = sum(torch.einsum('bhij,bhjd->bijd', chunk, z).norm(dim=-1).sum(1)
              for chunk in a.split(32, dim=2))
```

For the final-block intervention we hold the attention matrix fixed. This excerpt from the full-forward check removes one patch's term from the CLS query. The output projection retains its bias and the rest of the model runs normally.

```
# p: [batch, head, query, source]; v: [batch, head, source, head_dim]
head_out = p @ v
head_out[:, :, 0, :] -= p[:, :, 0, source_patch + 1, None] * v[:, :, source_patch + 1, :]
full = head_out.transpose(1, 2).reshape(batch, tokens, -1)
return attn.proj_drop(attn.proj(attn.norm(full)))
```

**AI Use Statement.** I set the direction of this project, chose a fast-weight perspective for studying ViT attention maps, and organized the paper around three assumptions about how these maps are interpreted. I used AI tools to help search the literature, design follow-up experiments, write code, run experiments, analyze results, and draft and revise the report.